

Chapter 4 and 5
Assignment 4

Deadline: March 3, 2026

Short Answer Questions:

1. Define Web Service.
2. What is the difference between SOAP and REST?
3. Write any two examples of HTTP methods.
4. What is an API endpoint?
5. Define WSDL.
6. What is meant by stateless communication?
7. Define Authentication.
8. Define Authorization.
9. What is HTTPS?
10. What is XSS attack?

Long Answer Questions:

1. Explain the working principle of a RESTful Web Service.
2. Explain the role of JSON in REST APIs.
3. What is API documentation? Why is it important?
4. Explain the difference between session-based authentication and token-based authentication.
5. What is SQL Injection? Explain with a simple example.
6. Explain CSRF attack in short.
7. What are secure cookies? Mention their types/flags.
8. Explain why input validation is important in web security.
9. Explain the working of **JWT authentication** with flow diagram.
10. Suppose you are building a Student Management API. Design 5 REST endpoints with methods and sample response format.
11. Why should sensitive data never be stored in localStorage?
12. Explain the difference between **encryption and hashing** with real examples.